Report 1: Project Description Report

Hospital Management System UML Class Diagram Assignment | ULK | October 2026

1. Project Title and Business Idea

Project Title: Hospital Management System (HMS)

Business Idea: An information system for a hospital that manages departments, doctors, patients, appointments, medical records, prescriptions, rooms and invoices in one centralised platform. The hospital registers patients, schedules appointments between patients and doctors, keeps a medical record for every patient, issues prescriptions, assigns rooms, and bills
patients through invoices.

2. Problem Statement

Hospitals deal with large amounts of data every day. With paper-based or manual processes, records get lost, appointments are double-booked, doctors cannot easily track a patient's history, and billing mistakes are common. The hospital needs an information system that organises its data and clearly models how patients, doctors, appointments, records, prescriptions, rooms and invoices relate to each other.

3. Purpose of the Proposed System

The purpose of the HMS is to: (1) centralise all hospital data, (2) make appointment scheduling fast and conflict-free, (3) give every patient a single medical record that doctors can update, (4) link prescriptions to appointments, and (5) automate patient invoicing. This assignment focuses on the analysis and design of the system using a UML class diagram.

4. Project Scope

The system WILL cover:

- Managing hospital departments and rooms.
- Registering and managing patient information.
- Managing doctors and their specialisations.
- Scheduling and tracking appointments between patients and doctors. Keeping one medical record per patient (diagnosis, treatment, history). Issuing prescriptions linked to appointments.
- Generating and paying invoices for patients.

The system WILL NOT cover:

- Developing a fully functional application (this project is a design exercise only).
- Integration with insurance companies or mobile-money payment gateways.
- Pharmacy stock / drug inventory management.
- Human-resource payroll for hospital staff.

5. Main Classes and Their Responsibilities

Hospital: Root of the system; contains departments; registers patients; produces reports.
Department: Logical unit of the hospital; employs doctors and has rooms.
Doctor: Examines patients, writes prescriptions, updates medical records. Patient: Registers at the hospital, books appointments, receives invoices.